

2 (a) State the principle of conservation of momentum.

When bodies in a system interact, the total momentum remains constant provided no resultant external force acts on the system.

.....
[1]

- (b)** Fig. 2.1 shows a metal bullet of mass 2.0 g fired horizontally into a block of wood of mass 600 g. The block is suspended from strings so that it is free to move in the vertical plane. The bullet hits and becomes embedded in the block. The block and bullet rise together through a vertical distance of 8.6 cm.

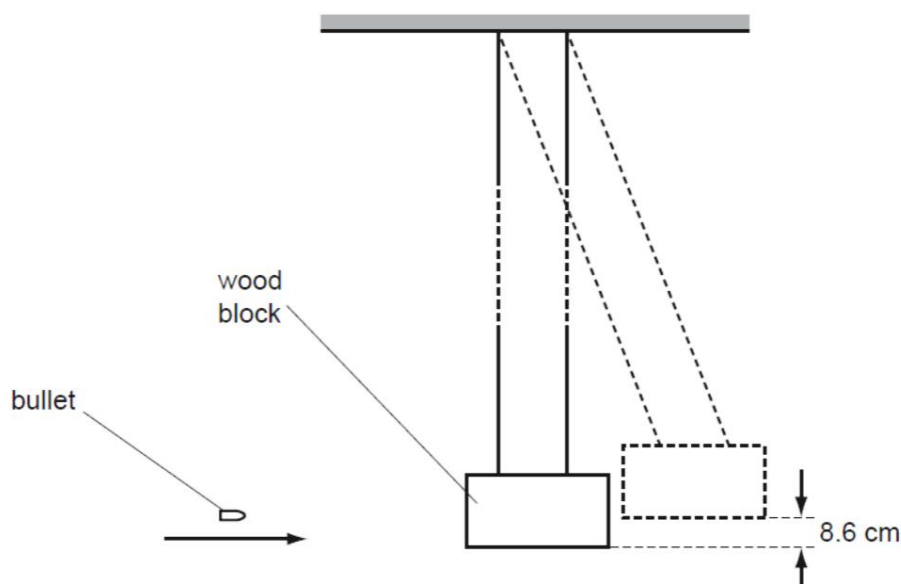


Fig. 2.1

- (i)** Show that the speed of the block and bullet as they just move off together is 1.3 m s^{-1} .

Comparing when bullet just become embedded in the block to highest position,

Gain in GPE = Loss in KE

$$(0.602)(9.81)(0.086) - 0 = KE_i - 0 \quad [\text{M1}]$$

$$v = \sqrt{\frac{(0.602)(9.81)(0.086) \times 2}{0.602}} \quad [\text{M1}]$$

$$= 1.3 \text{ m s}^{-1}$$

[2]

- (ii) Using (a) and (b)(i), determine the speed of the bullet before the impact with the block.

$$\text{PCOM, } \sum p_i = \sum p_f$$

$$m_{\text{bullet}} v_{\text{bullet}} + 0 = (m_{\text{bullet}} + m_{\text{block}}) \times v_{\text{bullet\&block}}$$

$$0.002 v_{\text{bullet}} = 0.602 \times 1.3 \text{ [M1]}$$

$$v_{\text{bullet}} = 391.3 \approx 390 \text{ ms}^{-1} \text{ [A1]}$$

speed = m s⁻¹ [2]

- (iii) A rubber bullet of the same mass hits the block with the same speed calculated in (ii) and rebounds in the opposite direction. State and explain whether the block will reach a maximum height of greater or less than 8.6 cm.

Change in momentum of bullet increase, (by PCOM or N3L) change in momentum of block increases too[B1].

Block will rise up with a greater speed/KE. Hence the maximum height of the block will be greater. [B1]

.....[2]

[Total : 7]

